

# ***Term Project: Instant Messaging App***

## **Feature Document**

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# **1 Introduction**

## **1.1 Purpose**

This document describes the audience, vision, scope, and features of a web-based instant messaging application designed for students.

## **1.2 Target Audience**

Students who want to communicate and share information related to their classes, study groups, and academic activities.

## **1.3 Terms and Definitions**

Channel – A group messaging space where multiple users can communicate with each other.

Chat – A one-on-one conversation between two users.

## 2 Product Overview

This product is a study-centric messaging app for students to better their grades.

**What:** An instant messaging app for students

**Who:** Students

**Why:** Improve their grades, or connect

### 2.1 Product Vision

**For:** Students

**Who:** Want to improve their grades, or connect with students

**The:** Instant messaging platform

**That:** Enables students to quickly share study resources and communicate academic information.

**Unlike:** Discord

**Our product:** an academic-focused instant messaging platform designed specifically for students to collaborate, ask questions, and share study resources in a structured, distraction-free environment.

### 2.2 Product Scope

#### 1. User Accounts & Authentication

- Sign up
  - Username + password
- Login / Logout
- Password reset
- Basic profile
  - Name
  - Profile photo (optional)
- Account deletion

## **2. Adding Users**

- Search users
  - By name
- Friend/connection requests
  - Send request
  - Accept / decline request
- Contact list
  - List of connected users

## **3. Channels (Group Communication)**

### **3.1 Making a Channel**

- Create channel
  - Channel name
- Channel type
  - Public
  - Private
- Invite users
  - Invite contacts
- Channel roles
  - Owner
  - Member

### **3.2 Channel Management**

- Leave channel
- Delete channel (owner only)
- Remove users (owner only)

## **4. Talking in Channels**

- Text messaging

- Message actions
  - Edit message
  - Delete message
- Search messages within the channel

## **5. Direct Messaging (Chats)**

### **5.1 Initiating a Chat**

- Start one-on-one chat from:
  - User profile
  - Contact list

### **5.2 Talking in Chats**

- Real-time text messaging
- Message actions
  - Edit
  - Delete

## **6. Blocking**

- Block users
  - Block from profile
- Effects of blocking
  - No direct messages allowed
- Unblock users

## **2.3 Operating Environment**

Web browser (Chrome, Firefox, Safari, Edge, etc.)

JavaScript enabled

## **2.4 Design and Implementation Constraints**

### **Database**

- MongoDB (Atlas)

### **Backend**

- Node.js + Express
- Socket.IO for real-time chat

### **Frontend**

- React + Axios

# 3 User Stories

This section describes what the users need and why. It explains how the product will support students in communicating, collaborating, and improving academic performance.

This section is necessary because knowing the users' needs ensures that the product solves real problems students face, features are prioritized based on real user value, design decisions are aligned with actual usage, and development focuses on what users truly need.

We should make enough personas to cover the wide variety of each usertype and enough stories to cover the key scenarios.

## 3.1 Personas

### Persona 1 - “Jeff the Struggler”

**Age:** 19

**Major:** Computer Science

**Goal:** Improve grades and understand difficult course material

**Behavior:** Often asks classmates for help, needs quick answers, hates long email threads

**Pain Points:**

- Hard to find classmates to study with
- Misses announcements or notes
- Doesn't want to use social apps for school

## **Persona 2 - “John the Golden Student”**

**Age:** 21

**Major:** Business

**Goal:** Manage group projects efficiently and easily

**Behavior:** Creates study groups, schedules meetings, shares documents

### **Pain Points:**

- Group chat gets messy
- Important messages get lost
- Hard to track who has seen tasks

## **Persona 2 - “Joel the New Student”**

**Age:** 18

**Major:** Engineering

**Goal:** Make new friends and find study partners

**Behavior:** Wants to join communities, connect with classmates

### **Pain Points:**

- Doesn't know who to contact
- Feels isolated
- Needs a safe way to connect with peers

## **3.2 User Scenarios**

### **3.2.1 <User Scenarios with Persona 1>**

#### **Jeff wants help with a homework problem**

He posts the question in a course channel and waits for replies.

#### **Jeff needs class notes quickly**

He searches the channel for the latest notes shared by classmates.

#### **Jeff wants to avoid distractions**

He uses the app instead of social media so he can stay focused.



### **3.2.2 <User Scenarios with Persona 2>**

#### **John creates a study group channel**

He invites group members and starts sharing project files.

#### **John wants to organize tasks**

He pins important messages or shares deadlines in the channel.

#### **John needs to remove a disruptive member**

He removes the user from the group channel.

### **3.2.3 <User Scenarios with Persona 3>**

#### **Joel wants to connect with classmates**

He searches users by name and sends connection requests.

#### **Joel wants to ask a class question privately**

He starts a direct chat with a classmate.

#### **Joel wants a safe environment**

He blocks users who behave inappropriately.

## **3.3 Epic User Stories**

Write at least three epic (Chapter 3 p28 3rd bullet point) user stories (Chapter 3 p27-29) with acceptance criteria in the following template. Try to define each small user story specifically so that they can be used in the sprint backlog.

### 3.3.1 <Epic User Stories 1>

|                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epic 1:</b> User Account Creation and Networking                                                                                                                                                           |
| <i>As a</i> new student<br><br><i>I want to</i> create an account, and some friends, and join a group<br><i>So that I can</i> connect and meet with peers                                                     |
| <b>Acceptance Criteria</b><br><i>Given</i> that a student is on the landing page<br><i>When</i> they provide a username and password and hit "Sign Up,"<br><i>Then</i> they can add friends and join channels |
| User story 1:<br>As a student, I want to create an account and log in so that my messages remain private.                                                                                                     |
| User story 2:<br>As a student, I want to search and add classmates by name so that I can find the specific people in my study group.                                                                          |
| User story 3:<br>As a student, I want to search for and join a public channel so I can connect with people in the same channel                                                                                |

### 3.3.2 <Epic User Stories 2>

|                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epic 2:</b> Organized Group Collaboration                                                                                                                                                                                                      |
| <i>As a</i> group leader, I want to<br><br><i>I want to</i> create and manage channels for specific courses or projects<br><i>So that I can</i> keep academic discussions organized and separate from general talk                                |
| <b>Acceptance Criteria</b><br><i>Given</i> a logged in user is on the main page,<br><i>When</i> they click "Create Channel" and invite specific contacts,<br><i>Then</i> a new messaging space is formed where only invited members can interact. |
| User story 1:<br>As a student, I want to create a public or private channel with a specific name so that I can categorize discussions by course number.                                                                                           |
| User story 2:<br>As a channel owner, I want to remove disruptive members from a group so that the study environment remains focused.                                                                                                              |
| User story 3:<br>As a student, I want to leave a channel once a project is finished so that my sidebar stays clean and relevant.                                                                                                                  |

### 3.3.3 <Epic User Stories 3>

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epic 3:</b> Real-Time Chatting                                                                                                                                                                                                                         |
| <p><i>As a</i> new student needing help,</p> <p><i>I want to</i> exchange real-time messages and manage my posted content,</p> <p><i>So that I can</i> get immediate answers to my questions and fix any mistakes I make in my shared notes.</p>          |
| <p><b>Acceptance Criteria</b></p> <p><i>Given</i> that a user is inside a channel or direct chat,</p> <p><i>When</i> they type a message and hit enter,</p> <p><i>Then</i> the message appears instantly for all participants without a page refresh.</p> |
| <p>User story 1:</p> <p>As a student, I want to send and receive real-time text messages so that I can have a fluid conversation.</p>                                                                                                                     |
| <p>User story 2:</p> <p>As a student, I want to edit or delete my sent messages so that I can correct typos in my shared study information.</p>                                                                                                           |
| <p>User story 3:</p> <p>As a student, I want to block a user who is being inappropriate so that I can maintain a safe and professional learning environment.</p>                                                                                          |

## 4 Feature List

|                                                                                                                   |                    |                           |
|-------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 1:</b> User Registration & Authentication                                                              | <b>Priority: 1</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>A system where students can sign up with a username/password and log in to access the app. |                    |                           |
| <b>Constraints:</b><br>Requires MongoDB for user info storage.                                                    |                    |                           |
| <b>Comments:</b><br>Needed for all other features                                                                 |                    |                           |

|                                                                                                                   |                    |                           |
|-------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 2:</b> Real-time Channel Messaging                                                                     | <b>Priority: 1</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>The ability to send and receive text messages in a group channel instantly using Socket.IO |                    |                           |
| <b>Constraints:</b><br>Requires a WebSocket connection, messages must be saved to the database.                   |                    |                           |
| <b>Comments:</b><br>This is the core "Product Vision" of the app                                                  |                    |                           |

|                                                                                                                   |                    |                           |
|-------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 3:</b> Contact Search & Friend Requests                                                                | <b>Priority: 1</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>The ability to send and receive text messages in a group channel instantly using Socket.IO |                    |                           |
| <b>Constraints:</b><br>A search bar to find peers by username and a simple "Send Request" button.                 |                    |                           |
| <b>Comments:</b><br>Key for connections                                                                           |                    |                           |

|                                                                                                                       |                    |                           |
|-----------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 4:</b> Channel Creation & Role Management                                                                  | <b>Priority: 2</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>Interface for creating Public/Private channels and assigning an "Owner" status to the creator. |                    |                           |
| <b>Constraints:</b><br>Only owners can delete the channel or remove members.                                          |                    |                           |
| <b>Comments:</b><br>Essential for John to manage his group projects.                                                  |                    |                           |

|                                                                                                               |                    |                           |
|---------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 5:</b> Message CRUD                                                                                | <b>Priority: 2</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>Users can click on their own sent messages to modify the text or remove them entirely. |                    |                           |
| <b>Constraints:</b><br>Users cannot edit or delete messages sent by others                                    |                    |                           |
| <b>Comments:</b><br>Helps Jeff fix mistakes in shared notes                                                   |                    |                           |

|                                                                                                        |                    |                           |
|--------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 6:</b> Direct Messaging                                                                     | <b>Priority: 2</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>Private chat functionality between two connected users, separate from channels. |                    |                           |
| <b>Constraints:</b><br>Must be restricted to "Friends" only.                                           |                    |                           |
| <b>Comments:</b><br>Good for private questions or one-on-one tutoring.                                 |                    |                           |

|                                                                                                              |                    |                           |
|--------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 7:</b> User Blocking System                                                                       | <b>Priority: 3</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>A "Block" button on user profiles to prevent direct messages and hide their presence. |                    |                           |
| <b>Constraints:</b><br>Blocked users should still be visible in shared public channels, but unable to DM.    |                    |                           |
| <b>Comments:</b><br>Crucial for safety and a "distraction-free" environment.                                 |                    |                           |

|                                                                                                 |                    |                           |
|-------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| <b>Feature 8:</b> Basic Profile Customization                                                   | <b>Priority: 3</b> | <b>Effort Required: 1</b> |
| <b>Description:</b><br>Users can update their display name and upload a profile photo.          |                    |                           |
| <b>Constraints:</b><br>Image storage might be needed, or just use URL links to keep effort low. |                    |                           |
| <b>Comments:</b><br>Purely aesthetic                                                            |                    |                           |